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Explainable Dynamic Graph Neural Networks for Predictive Maintenance in Vehicle Chassis Systems

Abstract

Predictive maintenance is essential for commercial vehicle fleets to reduce unexpected downtime and emergency repair costs. While standardized fault codes (SPN/FMI, representing *Suspect Parameter Numbers* and *Failure Mode Indicators*) assist in diagnosis, their temporal and spatial inconsistency limits the effectiveness of conventional time-series models in identifying high-cost failures. We propose a *Hybrid Node-level Relationship-based Graph Convolutional Network with Random Forest* (**NRP-GCN-RF**), which encodes fault interactions as graphs to capture non-temporal dependencies. Built on a real-world dataset from Dongfeng Motor Corporation, our study follows a dual-task design. (1) We construct a predictive model using graph neural networks (GCN) and random forests (RF) to forecast emergency repair costs and fault categories based on chassis-level fault sequences. (2) In parallel, we apply the Apriori algorithm to mine frequent co-occurring SPN-FMI pairs, revealing interpretable fault patterns and subsystem-level dependencies. This interpretable analysis complements the graph-based model by supporting feature design and failure diagnostics. Experiments show that our approach achieves **98.93%** accuracy, raises high-cost failure precision from **60%** to **95%**, and improves recall by **25%**, offering a robust and explainable solution for predictive maintenance in commercial fleets.

Keywords: Fault Prediction, Predictive Maintenance, Graph Neural Networks,Machine Learning, Association Rule Mining, Vehicle Maintenance, Explainable AI

1. Introduction

Commercial vehicle fleets face increasing demands for reliability, safety, and cost efficiency. Unexpected mechanical failures—particularly those requiring emergency roadside repairs—can lead to substantial downtime, safety risks, and unplanned operational expenses. Although modern vehicles continuously record diagnostic fault codes through onboard monitoring systems, these fault logs are typically sparse, irregular, and highly heterogeneous. Such data characteristics make traditional sequence-based or rule-based fault prediction models unreliable for large-scale fleet maintenance.

Recent advances in graph neural networks (GNNs) have opened new possibilities for modeling relational dependencies in complex industrial systems. In the context of commercial vehicle diagnostics, fault codes rarely occur independently; they often emerge as interrelated patterns within the same vehicle or across similar operating conditions. These patterns may reflect temporal propagation, spatial proximity, or subsystem interdependence. Capturing these cross-fault and cross-chassis relationships is crucial for both improving predictive accuracy and enhancing engineering interpretability.

To address these challenges, we propose a hybrid learning framework that combines directed graph construction, GCN-based feature learning, and ensemble classification. Raw fault sequences are transformed into multi-scale graph structures, where nodes represent chassis-fault instances and edges encode statistical, spatial, and temporal relationships.

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The GCN aggregates information from both local and global graph neighborhoods to capture fault propagation dynamics and latent hub patterns that conventional models often overlook. The resulting graph embeddings are further processed using a Random Forest classifier, providing robust predictions even under severe class imbalance.

To comprehensively evaluate this framework, we define three prediction tasks that reflect real-world maintenance requirements in commercial fleets: (i) predicting whether an emergency repair fee was incurred (binary classification), (ii) classifying the repair cost into four categories (multi-class), and (iii) identifying the fault type among nine categories derived from NLP-based clustering of fault descriptions. Beyond predictive accuracy, our framework emphasizes interpretability and engineering relevance. By integrating Apriori-based association rule mining, we uncover frequent co-fault motifs that provide transparent reasoning about subsystem dependencies and actionable insights for diagnosis and scheduling. Unlike prior work focused only on temporal modeling or individual code analysis, our approach combines graph topology, operational context, and interpretable rules to form a unified, explainable solution for predictive maintenance.

The main contributions of this study are threefold. First, we propose a decoupled pipeline integrating graph representation, GCN-based feature learning, and Random Forest classification, which achieves robust performance under severe class imbalance while preserving interpretability. Second, we construct directed, weighted graphs using spatial, temporal, and co-occurrence attributes, and validate them with interpretable association rules that provide diagnostic evidence. Third, we demonstrate effectiveness on multi-year industrial data from Dongfeng Motor Corporation, achieving nearly 99% overall accuracy and significant gains in precision and recall for high-cost faults, confirming both technical and practical value.

2. Related Work

Research on vehicle fault detection spans three principal methodological directions: statistical approaches, sequence modeling, and graph-based frameworks (Zhang et al., 2020, 2019; Connor et al., 1994; Wen et al., 2023; Jin et al., 2023, 2022; Biddle and Fallah, 2021; Theissler, 2017; Choi et al., 2021; Namburu et al., 2019; Moura et al., 2011; Zheng and Tong, 2023; Ifeanyi, 2024; Wang et al., 2022). Each of these paradigms targets different aspects of the complex, multi-modal data underlying commercial vehicle operation and fault occurrence.

Statistical methods, such as frequency analysis and association rule mining, have been widely used to identify frequently occurring or co-occurring fault codes in large datasets (Zhang et al., 2020, 2019). For example, sliding window analysis and threshold-based filtering are employed to monitor changes in fault patterns over time. While these techniques can provide fast, interpretable results, they are fundamentally limited by assumptions of stationarity and regularity in the underlying data (Theissler, 2017). In practical deployments, vehicle operational environments and fault occurrence patterns are dynamic and noisy, often leading to unreliable performance when using static or hand-crafted thresholds. Additionally, statistical approaches typically perform poorly in detecting rare but high-impact failures and may be sensitive to missing or erroneous data.

Sequence modeling approaches leverage temporal dependencies between fault events to provide more context-aware prediction and detection. Methods based on recurrent neural networks (RNNs), gated recurrent units (GRUs), and Transformer architectures have demonstrated improvements in tracking evolving fault sequences and extracting semantic relationships among diagnostic codes (Connor et al., 1994; Wen et al., 2023). For instance, GRU-based frameworks capture temporal progression of faults across usage cycles, and embedding-based methods utilize models such as Word2Vec and Transformers to uncover latent associations in code sequences. However, these models are often limited in their ability to represent long-range or non-adjacent dependencies, and may not adequately capture global relationships between system components or across multiple fault events.

Graph-based frameworks have recently gained traction for their capacity to explicitly model interactions and dependencies within complex systems (Jin et al., 2023, 2022). Heterogeneous and dynamic graph neural networks (GNNs) allow for the integration of spatial, operational, and relational information, improving the expressiveness and flexibility of predictive models. Despite these advances, most graph-based studies are restricted to static graphs or address only single-task prediction problems, limiting their ability to generalize across multiple failure types or operational contexts (Biddle and Fallah, 2021; Theissler, 2017; Choi et al., 2021; Namburu et al., 2019). The challenges of multi-fault interaction, transfer learning across vehicle platforms, and scalability in real-world settings are still largely unsolved.

Efforts to incorporate **adaptive thresholding** and real-time adjustment mechanisms have further improved model robustness under changing operational conditions (Moura et al., 2011; Namburu et al., 2019). However, these solutions are rarely unified with advanced sequence or graph learning approaches, and thus often fail to deliver a comprehensive answer to the joint challenges of explainability, class imbalance, and dynamic data distributions.

In summary, while significant progress has been made using statistical, sequential, and graph-based techniques, few existing works offer a holistic solution that jointly addresses dynamic, multi-fault, cross-platform, and explainable predictive maintenance for vehicle systems. This motivates our proposed framework, which integrates dynamic GNNs, ensemble learning, and adaptive thresholding in a unified, multi-task setting.

3. Problem Formulation & Tasks

Data and notation. Let $\mathcal{D} = \{(\mathcal{S}_i, \mathbf{z}_i)\}_{i=1}^{N}$ be a fleet dataset, where each vehicle i has a time-ordered fault sequence $\mathcal{S}_i = \langle (s_{i1}, \tau_{i1}), (s_{i2}, \tau_{i2}), \ldots \rangle$, with diagnostic code s_{ij} (e.g., SPN–FMI) and timestamp τ_{ij}. Optionally, $\mathbf{z}_i$ aggregates vehicle-level statistics (usage, region, etc.). We construct a directed attributed graph

$$G = (V, E, X), \qquad V = \{\, v = (c, s, \tau) \,\}, \quad X \in \mathbb{R}^{|V| \times d},$$

where each node v binds a chassis c, a code s, and an event time τ; X stacks node features (temporal, spatial, statistical). Edges E encode (i) temporal adjacency within a chassis, (ii) spatial proximity across chassis, and (iii) co-occurrence relations mined from maintenance logs. Their weights are fused as

$$w_{uv} = \alpha\, w_{uv}^{\text{temp}} \;+\; \beta\, w_{uv}^{\text{spat}} \;+\; \gamma\, w_{uv}^{\text{cooc}}, \quad \alpha, \beta, \gamma \geq 0.$$

Learning objectives. Given G, an encoder F_θ (e.g., a GNN) produces node embeddings $\mathbf{H} = F_\theta(G) \in \mathbb{R}^{|V|\times d}$ and a pooled graph embedding $\mathbf{g} = \mathrm{Pool}(\mathbf{H})$. We optimize three heads:

Y1 (binary, Has-Emergency-Fee). Predict whether a vehicle requires emergency fee:

$$\hat{y}_1 = f_\theta(G) \in \{0, 1\}, \qquad \mathcal{L}_1 = \mathrm{BCE}(y_1, \hat{y}_1).$$

Y2 (4-class, Emergency-Fee-Category). Predict cost level:

$$\hat{y}_2 = g_\theta(G) \in \{0, 1, 2, 3\}, \qquad \mathcal{L}_2 = \mathrm{CE}(y_2, \hat{y}_2).$$

Y3 (9-class node classification, Fault-Category). For each node $v \in V$, predict its subsystem/category:

$$\hat{y}_3(v) = h_\theta(v \mid G) \in \{1, \ldots, 9\}, \qquad \mathcal{L}_3 = \frac{1}{|V|} \sum_{v\in V} \mathrm{CE}\big(y_3(v), \hat{y}_3(v)\big).$$

The total loss is a weighted sum

$$\mathcal{L} = \lambda_1\mathcal{L}_1 + \lambda_2\mathcal{L}_2 + \lambda_3\mathcal{L}_3, \qquad \lambda_1, \lambda_2, \lambda_3 \geq 0.$$

In our framework, the notion of *dynamic* is not limited to static snapshots of fault logs. Graphs are constructed by respecting the temporal order of fault events, so that edges emerge naturally from event precedence. Moreover, the design supports incremental extension: when new faults arrive, additional nodes and edges can be inserted through threshold-based edge creation and cached neighborhoods. In this paper we evaluate the framework on historical replay windows, while the same mechanism can be directly applied in online settings. This interpretation keeps the title consistent with our design intent and clarifies that "dynamic" refers to both time-aware construction and the potential for real-time updates.

4. Data Labeling

This section presents a comprehensive overview of the data used in this study, including preprocessing steps, task-specific label construction, and the mining of co-occurring fault patterns. We first describe the structure and cleaning of fault and maintenance records, followed by dataset alignment and labeling for multi-task prediction. Finally, we analyze frequent SPN-FMI combinations to support downstream graph construction.

4.1. Dataset Overview and Preprocessing

We utilized two proprietary datasets provided by Dongfeng Motor Corporation, one of China's leading commercial vehicle manufacturers. These real-world datasets were collected from Dongfeng's national fleet management system. Specifically, the data consist of an automatically generated fault log dataset and a manually recorded maintenance dataset containing repair information and emergency fees. As genuine industrial data rather than simulated samples, they capture authentic operating conditions, diverse fault patterns, and

realistic cost behaviors, providing a rare and valuable foundation for research on predictive maintenance in large-scale vehicle fleets. Due to confidentiality and data protection agreements, the raw datasets are not publicly available. However, interested parties may contact Dongfeng Motor Corporation for potential access or collaboration under appropriate data-sharing arrangements.

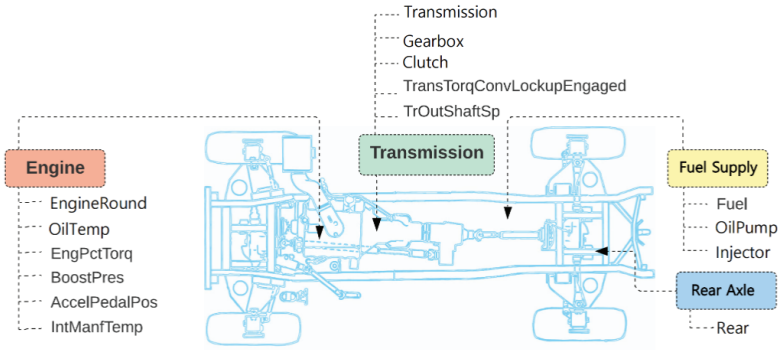


Figure 1: Reference diagram of automotive systems and fault categories.

Table 1: Fault clustering rules based on keywords

No.	Keywords in Fault Description
1	Rear Axle, Axle
2	Engine, Oil
...	...
8	ECU, Electronics, Motor
9	Other Fault Descriptions

The fault dataset includes fields such as *ChassisNumber, SystemType, FaultName, FaultCount, SPN, FMI, FaultStartTime*, and *FaultEndTime*. While *FaultName* has over 700 unique entries, only 135 distinct *SPN-FMI* pairs are observed, providing a more compact and standardized representation of faults. To improve consistency, fault records were grouped by *ChassisNumber* and sorted by *FaultStartTime* to form vehicle-specific sequences. Rare *SPN-FMI* pairs below a frequency threshold were excluded to reduce noise and focus on representative patterns.

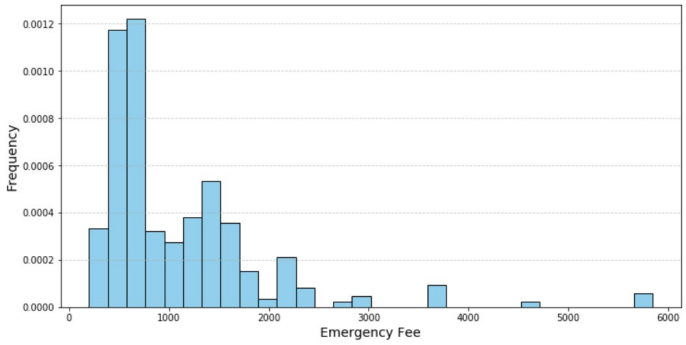


Figure 2: Frequency distribution of emergency fees (continuous scale).

The maintenance dataset includes fields such as *FaultDate, CheckDate, FaultCode, FaultDescription, PartName*, and *EmergencyFee*. Since the descriptions span over 200 unique fault types, we applied a keyword-based NLP approach to cluster them into nine broader fault categories aligned with automotive subsystem references. Table 1 summarizes the key-

word mapping rules, and Figure 1 illustrates the association between system components and their respective categories.

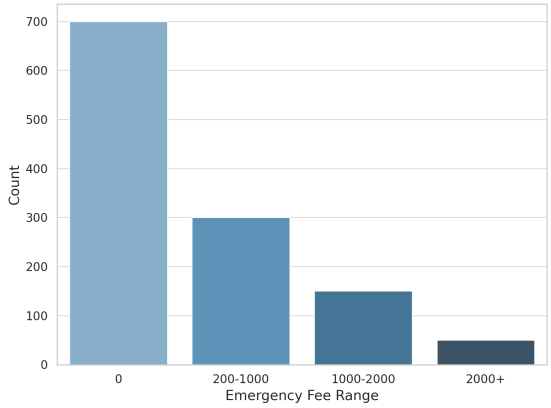


Figure 3: Categorical distribution of emergency fee ranges.

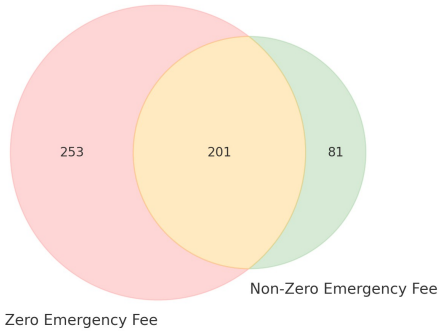


Figure 4: Venn diagram of SPN-FMI combinations associated with zero and non-zero emergency fees.

To support emergency cost prediction, we analyzed the distribution of emergency fees. Figure 2 shows the continuous distribution, revealing that most fees fall below 2,000 units, with a clear peak under 1,000. Based on this, we discretized the fees into four predefined brackets: **0–200**, **200–1,000**, **1,000–2,000**, and **above 2,000**. The corresponding categorical distribution is illustrated in Figure 3.

To further investigate cost-driving patterns, we analyzed SPN-FMI combinations by whether they co-occur with zero or non-zero emergency fees. As shown in Figure 4, 201 fault codes appear in both categories, while 253 and 81 are exclusive to zero and non-zero fee cases, respectively. This overlap suggests that some structural faults may manifest across a range of cost severities. These insights lay a foundation for cost-aware fault prediction.

4.2. Dataset Alignment and Task Labeling

To integrate the fault and maintenance datasets, entries were matched based on *Chassis-Number* and relevant date fields. By aligning *FaultStartTime* from the fault dataset with *FaultDate* in the maintenance records, a direct connection was established between early fault occurrences and subsequent repairs. This integration facilitates the three core prediction tasks mentioned in the introduction.

4.3. Mining Frequent Fault Combinations

Upon integrating the datasets, a key observation was the significant variation in the occurrence of *SPN-FMI* pairs across different vehicles. For example, the *523004-16* combination appears in over 50% of fault sequences for certain chassis, whereas *792-2* is present in only 20% of cases. This disparity suggests that some fault combinations may be more predictive of future failures or higher repair costs.

Table 2: Classification of SPN-FMI Occurrence Ranges

Range	Proportion	Characteristics
Low	0–10%	Stable associations with minimal variation
Medium	10–25%	Moderate variation, context-dependent patterns
High	>25%	Significant fluctuation, potential transient dependencies

To analyze co-occurring fault patterns, we employed an optimized *Apriori* algorithm with five steps: (1) filtering low-frequency SPN-FMI pairs; (2) constructing a sparse boolean matrix representing chassis–fault relationships; (3) compressing rows and columns to reduce redundancy; (4) mining frequent itemsets using a 1% support threshold; and (5) indexing patterns with a trie structure for efficient retrieval.

The resulting combinations were categorized into three stability ranges, as shown in Table 2. By combining matrix compression with trie-based indexing, the approach improves computational efficiency while preserving mining accuracy. The extracted patterns support early warnings and resource allocation in predictive maintenance.

5. Model Design and Graph Construction

We evaluated three modeling approaches for vehicle fault prediction: Support Vector Machines (SVM), Random Forests (RF), and our proposed hybrid model combining Graph Neural Networks (GNNs) with Random Forests (NRP-GCN-RF). The first two serve as baselines, highlighting the limitations of conventional machine learning on sparse, high-dimensional data. We then present the design of NRP-GCN-RF, which integrates graph-based relational learning with ensemble classification to capture both temporal and non-temporal dependencies in complex fleet fault data.

5.1. Support Vector Machines (SVM)

Table 3: Simulated dataset showing the "large p, small n" problem with high-dimensional sparse features.

Chassis	FaultLabel	523004-16	520415-0	792-2	523014-8	...
K1036118	1	1	1	0	0	...
K1036118	2	1	1	0	0	...
L8043320	1	0	0	1	1	...
L8043320	2	0	0	1	1	...
L8058404	1	0	0	0	0	...

SVM models were trained on one-hot encoded SPN–FMI fault codes, yielding sparse binary matrices with tens of thousands of features. This creates the "large p, small n" problem, where the feature dimension greatly exceeds the number of samples, and overlap across vehicles is minimal. Inner-product–based kernels fail to capture meaningful similarities, leading to underfitting. Limited improvements were observed when applying feature

selection based on frequent co-occurrence combinations, but SVM could not model relational dependencies or long-range interactions among fault codes.

5.2. Random Forests (RF)

Table 4: Restructured Fault Dataset with Tabular Attributes

ChassisNumber	FaultLabel	High Proportion Combinations	Counts	AvgLongitude	AvgLatitude	...
K1036118	1	['523004-16', '520415-0']	[5, 5]	117.2451	39.3165	...
K1036118	2	['523004-16', '520415-0']	[7, 6]	117.2066	39.3501	...
		...				
L8043320	1	['792-2', '523014-8', ...]	[14, 14, ...]	122.9887	41.0847	...
L8058404	1	['518110-5', '518118-5', ...]	[14, 13, ...]	103.0202	43.2948	...

RF models were trained on tabular features including frequent combinations, counts, proportions, and geospatial attributes. While RF captured structured patterns and achieved solid baseline performance, it struggled with rare but high-cost failures due to class imbalance. To mitigate this, we applied SMOTE to synthesize new minority samples:

$$\hat{x}_{\text{new}} = x_{\text{minority}} + \lambda \cdot (x_{\text{neighbor}} - x_{\text{minority}}), \quad \lambda \sim \text{Uniform}(0, 1).$$

This improved minority recall, but RF alone could not model temporal sequences or inter-chassis dependencies.

5.3. Proposed Hybrid Model: NRP-GCN-RF

To overcome these limitations, we propose NRP-GCN-RF, a hybrid framework that integrates Graph Neural Networks (GNNs) with Random Forests (RF). This approach encodes vehicle faults as graphs, enabling both temporal and non-temporal dependencies to be modeled across the fleet.

Graph Construction Raw logs are segmented into time-ordered sequences per chassis. We build a directed attributed graph

$$G = (V, E, X), \qquad V = \{\, v = (c, s, \tau) \,\}, \quad X \in \mathbb{R}^{|V| \times d},$$

where each node v binds a chassis c, a diagnostic code s (e.g., SPN–FMI), and an event time τ; X stacks temporal/spatial/statistical node features. Edges capture three types of relations:

- **Temporal adjacency** (w^{temp}): within-chassis precedence/overlap between consecutive events inside a rolling window Δt.
- **Spatial proximity** (w^{spat}): cross-chassis co-location within a radius Δs (e.g., 0.05° by lat/lon or r km by Haversine).
- **Co-occurrence** (w^{cooc}): high-support pairs from Apriori mining over maintenance logs, with minimum support $\sigma \geq 1\%$ (confidence/lift optionally used for reweighting).

The final edge weight fuses these signals:

$$w_{ij} = \alpha\, w_{ij}^{\text{temp}} + \beta\, w_{ij}^{\text{spat}} + \gamma\, w_{ij}^{\text{cooc}}, \qquad \alpha, \beta, \gamma \geq 0,$$

where α, β, γ control the contribution of temporal/spatial/statistical relations.

Sparsification prior to training. Before any GPU stage, we apply a three-step pruning to control density: (i) keep only time-near pairs within Δt; (ii) keep only spatial neighbors within Δs; (iii) drop weak edges with $w_{ij} < \varepsilon$ after fusion. Let $V_{\text{win}}, E_{\text{win}}$ denote nodes/edges in a (chassis, time-window) shard. We then *parallelize* construction across shards (multi-process I/O and batched edge creation), so the peak host memory per worker scales as $\tilde{O}(|V_{\text{win}}| + |E_{\text{win}}|)$. Empirically, after sparsification we obtain $|E| \approx c\,|V|$ with a small constant c (bounded average degree), which yields near-linear construction time.

5.3.1. Scalability & Resource Usage

Mini-batch GNN propagation. We train on subgraphs with neighborhood sampling. For minibatch size B, hidden width d, per-hop neighbor cap $\bar{k}$, and K layers, message passing costs

$$T_{\text{GNN}} = O\big(K\, B\, d\, \bar{k}\big), \qquad \mathsf{Mem}_{\text{GPU}} \approx O(B\, d) + O\big(B\, \bar{k}\big),$$

covering activations and sampled adjacency. Capping $\bar{k}$ bounds both time and memory; sequentially processing (chassis, window) shards further limits the end-to-end peak footprint to the largest in-memory subgraph.

Apriori at scale (with FP-Growth fallback). The worst-case candidate growth up to length L over M items is $\sum_{k=1}^{L} \binom{M}{k}$. We keep mining tractable by: (i) setting a minimum support $\sigma \geq 1\%$; (ii) *rare-binning* infrequent SPN–FMI codes into an `<other>` bucket before mining; (iii) row/column compression and trie indexing to avoid redundant scans. Under extreme sparsity or long-tailed codes, we optionally switch to FP-Growth to avoid breadth-first candidate generation. These controls address scalability/real-time concerns without introducing new experiments.

5.4. Proposed Hybrid Model: NRP-GCN-RF

Motivation and Contributions. We propose **NRP-GCN-RF**, a hybrid framework that (i) encodes fleet fault logs as a directed, attributed graph and (ii) *decouples* relational representation learning (GNN) from classification (Random Forest). Compared with conventional tabular/sequence models, NRP-GCN-RF explicitly fuses *temporal* dependencies (within-chassis event order) and *non-temporal* relations (spatial proximity, cross-chassis co-occurrence), improving high-cost failure detection and interpretability. Our novelty lies in: (1) a *multi-factor* edge design grounded in operational semantics (temporal, spatial, association); (2) an *encoder-agnostic* graph backbone with a decoupled RF head for robustness and feature importance; and (3) *actionable* explanations via Apriori rules and graph visualizations. Section **??** provides GIN/Graph-Transformer comparisons; Section **??** analyzes temporal vs. non-temporal contributions.

Problem Definition. Let the raw logs of chassis c be a time-ordered sequence $\mathcal{S}_c = \{(s_t, \tau_t, \ell_t)\}_{t=1}^{T_c}$, where s_t is an SPN–FMI code, τ_t the timestamp, and ℓ_t the location. We build a directed graph

$$G = (V, E, X), \quad V = \{v_i = (c_i, s_i, \tau_i)\}, \quad X \in \mathbb{R}^{|V| \times d},$$

where X aggregates code identity embeddings, temporal lags (e.g., time-since-last-fault), chassis metadata, and spatial statistics. We consider:

$$\textbf{HasEmergencyFee:} \quad f_G : \mathcal{G} \to \{0,1\}, \quad \hat{y} = \mathbb{1}(\Pr(y{=}1 \mid G) > \theta), \tag{5.1}$$

$$\textbf{EmergencyFeeCategory:} \quad g_G : \mathcal{G} \to \{1, \ldots, C\}, \tag{5.2}$$

$$\textbf{FaultCategory (node-level):} \quad h_V : \mathcal{V} \to \{1, \ldots, K\}, \tag{5.3}$$

where $\mathcal{G}$ is the set of graphs (per chassis window or mini-batch subgraph), $\mathcal{V}$ nodes within G, C cost bins, K subsystem categories, and θ an adaptive threshold.

From Sequences to Graphs. As illustrated in Figure 5, we segment logs by chassis, sort by *FaultStartTime*, and project into a unified graph. Each node encodes a *chassis–fault–timestamp* triple; edges capture temporal, spatial, and statistical relations, forming a *hypervariate fault graph* that preserves intra-chassis order and inter-chassis similarity.

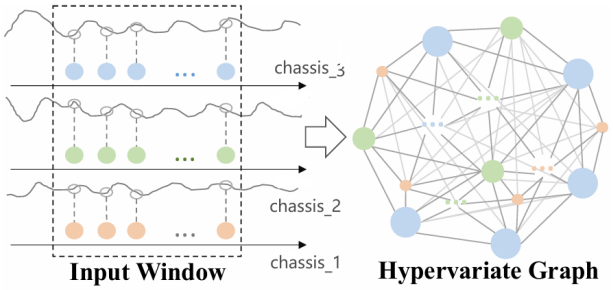


Figure 5: Graph Conversion

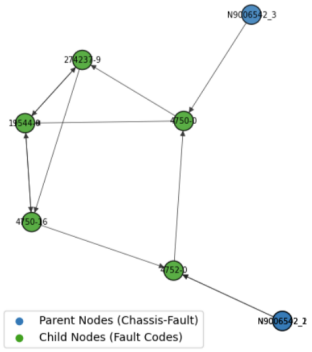


Figure 6: Directed Subgraph

Figure 5 shows how time-sorted sequences become graphs whose nodes are chassis–fault instances and whose edges span intra-/inter-chassis relations. Figure 6 zooms into a chassis subgraph: blue nodes are chassis IDs, green nodes are SPN–FMI codes, and arrows encode temporal order.

Edge Criteria and Multi-Factor Weights. A directed edge $i \to j$ is created if *any* of the following holds:

1. **Temporal adjacency:** events are consecutive or overlap within window Δ_t.
2. **Spatial proximity:** geodesic distance $d_{\text{geo}}(\ell_i, \ell_j) \le \Delta_s$ (e.g., 0.05°).
3. **Statistical co-occurrence:** (s_i, s_j) passes Apriori thresholds (support/confidence).

We fuse heterogeneous evidence as

$$w_{ij} = \alpha\,\mathrm{sim}_t(v_i, v_j) + \beta\,\mathrm{sim}_s(v_i, v_j) + \gamma\,\mathrm{assoc}(v_i, v_j), \qquad \alpha, \beta, \gamma \geq 0, \quad \alpha+\beta+\gamma = 1, \quad (5.4)$$

where sim_t is a decayed temporal kernel, sim_s a spatial kernel (e.g., an RBF of d_{geo}), and assoc a normalized Apriori statistic (confidence/lift). We retain edges with $w_{ij} \geq \varepsilon$ and row-normalize to obtain $\tilde{A}$.

Dynamic Aspect. "Dynamic'' denotes that G evolves as logs arrive: when a new event appears, we add v_{new} and recompute incident edges via (5.4). Although our experiments use historical windows, the construction and inference support online updates (Sec. **??**).

Encoder: Backbone-Agnostic GNN. We adopt a multi-layer GCN to learn node embeddings $H = \mathrm{GCN}(X, \tilde{A})$ via

$$h_v^{(k)} \;=\; \sigma\left(\sum_{u \in \mathcal{N}(v)} \frac{1}{\sqrt{d_v d_u}}\, W^{(k)} h_u^{(k-1)}\right), \qquad h^{(0)} = X, \quad (5.5)$$

where $W^{(k)}$ are trainable weights and σ is a nonlinearity. The framework is *encoder-agnostic*: GCN can be replaced by GIN/GAT/Graph Transformers under a fixed split/tuning budget.

Decoupled RF Heads. We decouple representation learning from decision making. For graph-level tasks we pool node embeddings (mean/max/attention) into z_G; for node-level we use per-node h_v:

$$\hat{y}^{\mathrm{graph}} = \mathrm{RF}(z_G), \qquad \hat{y}^{\mathrm{node}} = \mathrm{RF}(h_v).$$

Decoupling enhances robustness to class imbalance/noise and yields feature importance for diagnostics. For severe imbalance (e.g., high-cost failures) we optionally apply SMOTE in the embedding space and/or class-weighted sampling (Sec. **??**).

Feature Enrichment and Interpretability. Operational (duty cycle, time-since-last-fault), spatial, and Apriori features are fused at node/edge levels. Local clusters (Figure **??**) and the global fault graph (Figure 7) expose bridges/hubs across subsystems, aligning human-understandable rules with model predictions.

End-to-End Workflow. Figure 8 summarizes the pipeline: *preprocessing* → *SMOTE (optional)* → *graph construction* → *feature extraction* → *GNN training* → *transfer learning* → *fine-tuning* → *anomaly detection* → *adaptive thresholding*.

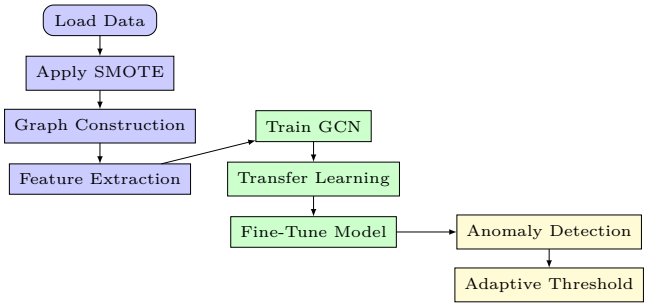


Figure 8: Workflow Overview

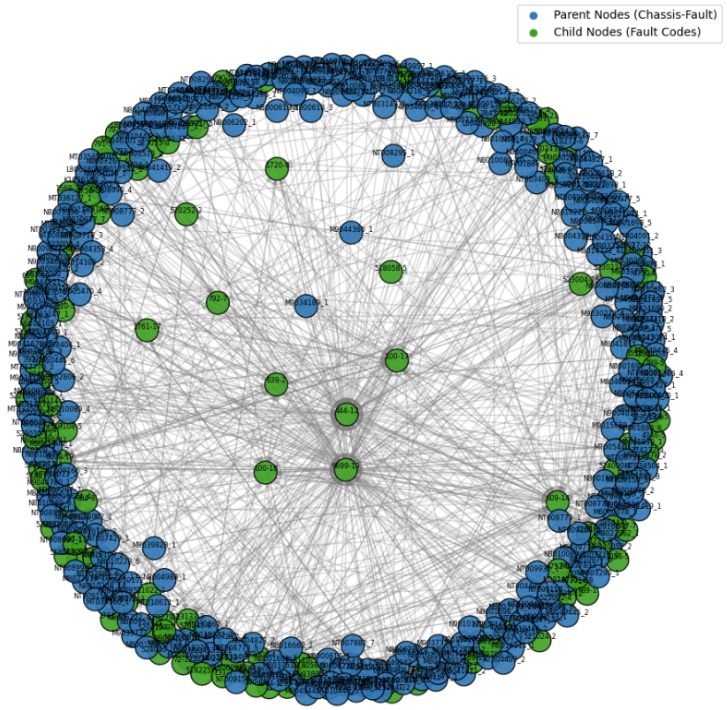


Figure 7: Global Graph

Complexity (Sketch) and Scalability. Graph construction is $O(|\mathcal{R}|)$ with parallel preprocessing over records $|\mathcal{R}|$; Apriori is $O(F(K, L))$ with K frequent items and max rule length L (bounded by support). Training with neighborhood sampling is $O(B\, d\, K_{\text{msg}})$ per batch (batch size B, feature dim d, effective neighbors K_{msg}). We further control memory/time via sparsification thresholds $(\Delta_t, \Delta_s, \varepsilon)$ and mini-batch subgraphing (Sec. 5.3.1).

Planned Analyses. To address reviewer concerns, Sec. **??** reports GIN and Graph Transformer backbones under identical splits/budgets; Sec. **??** decomposes temporal vs. non-temporal effects by (i) pure temporal GRU/LSTM baselines, (ii) reduced temporal weighting (α=0 in (5.4)), and (iii) full multi-factor edges, clarifying contribution ratios.

Summary. NRP-GCN-RF provides (i) a principled multi-factor graph abstraction of fleet faults, (ii) an encoder-agnostic GNN with a robust, interpretable RF head, and (iii) actionable rule- and graph-based explanations for predictive maintenance.

6. Experiments and Evaluation

We empirically assess NRP-GCN-RF on three prediction tasks: Y_1 **HasEmergencyFee** (binary, graph-level), Y_2 **EmergencyFeeCategory** (multi-class, graph-level), and Y_3 **FaultCategory** (multi-class, node-level). Unless otherwise stated, we adopt *vehicle-wise, chronological* K-fold cross-validation to prevent temporal and inter-vehicle leakage. We report standard classification metrics and provide ROC/PR curves and confusion matrices for key settings (Appendix).

Evaluation protocol. Hyperparameters (GNN depth/width and the edge-weight coefficients α, β, γ in Eq. (5.4)) are tuned on validation folds. For Y_1 we report AUROC, AUPRC, F1 and Precision@Top-k (risk ranking); for Y_2 we report Macro-/Weighted-F1 and quadratic weighted kappa (QWK); for Y_3 we report Micro-/Macro-F1 and per-class recall. We additionally examine calibration (ECE) and operating-point selection under cost constraints (Appendix).

6.1. Fault Code Association Analysis

To complement supervised prediction with interpretable structure, we apply Apriori to mine frequently co-occurring SPN–FMI pairs. These rules inform graph edge design (Sec. 5.4) and support subsystem-level diagnostics.

6.1.1. Apriori overview

Given a transaction set T (per-chassis windows), Apriori extracts frequent itemsets whose support exceeds a threshold and derives rules $A \Rightarrow B$ satisfying a minimum confidence:

$$\mathrm{supp}(A) = \frac{|T_A|}{|T|}, \qquad \mathrm{lift}(A \Rightarrow B) = \frac{\mathrm{supp}(A \cup B)}{\mathrm{supp}(A)\,\mathrm{supp}(B)}.$$

To curb spurious high lift on rare items, we enforce domain-informed minimum support and verify rule stability across folds (Appendix).

6.1.2. Strongly associated rules

Table 6 summarizes representative high-lift rules with perfect confidence. For example, (1761-18) ⇒ (1241-1) achieves lift 98, indicating a strong functional linkage.

Table 5: Association rules discovered by Apriori.

Table 6: Strongly associated rules.

Antecedents	Consequents	Support	Confidence	Lift
(1761-18)	(1241-1)	0.0102	1.00	98.0
(520243-0)	(639-5)	0.0102	1.00	98.0
(523053-8)	(523056-8)	0.0204	1.00	49.0
(518112-5)	(131-7)	0.0204	1.00	49.0
(4794-1)	(523006-4)	0.0102	1.00	32.7

Table 7: Moderately associated rules.

Antecedents	Consequents	Support	Confidence	Lift
(524040-20)	(597-13)	0.0196	1.00	51.0
(518108-5)	(790-5)	0.0196	1.00	34.0
(521022-2)	(520252-2)	0.0588	0.86	14.6
(1720-9)	(1624-2)	0.0882	0.90	10.2
(518114-5)	(4750-16)	0.0294	0.75	6.38

Insight. High-lift pairs imply subsystem interdependencies (e.g., EECU with induction), while moderate-lift cross-module rules highlight candidates for targeted inspection. These patterns enhance interpretability beyond black-box prediction.

6.2. Task-wise Performance with NRP-GCN-RF

We next report task-specific results for NRP-GCN-RF. The tables below retain the original class-wise metrics for clarity and traceability.

Y_1**: HasEmergencyFee.** The model achieves high recall on the positive class, indicating effective early warning of emergency repairs (Table 8).

Table 8: Classification report for Y_1: *HasEmergencyFee*.

Class	Precision	Recall	F1-Score	Support
0	0.91	0.84	0.87	25
1	0.94	0.97	0.96	67
Accuracy		0.9348 (on 92 samples)		
Macro Avg	0.93	0.91	0.92	92
Weighted Avg	0.93	0.93	0.93	92

Table 9: Classification report for Y_2: *EmergencyFeeCategory*.

Class	Precision	Recall	F1-Score	Support
0.0	1.00	0.82	0.90	17
1.0	0.92	0.94	0.93	81
2.0	0.96	0.93	0.95	86
3.0	0.95	1.00	0.97	73
Accuracy		0.9455 (on 257 samples)		
Macro Avg	0.96	0.92	0.94	257
Weighted Avg	0.95	0.95	0.95	257

Y_2**: EmergencyFeeCategory.** All bins exhibit strong F1, with perfect recall for the highest-cost bin (Class 3.0) and perfect precision for the lowest-cost bin (Class 0.0), indicating low false alarms for low-cost events and high sensitivity for costly cases.

Y_3**: FaultCategory.** Performance is consistently high across nine subsystems (Tables 10 and 11), with overall accuracy 0.989. Despite small support, rare classes remain serviceable.

Table 10: Classification report for Y_3: *FaultCategory* (subset).

Class	Precision	Recall	F1-Score	Support
0	0.96	1.00	0.98	50
1	1.00	0.67	0.80	3
2	1.00	0.98	0.99	62
3	0.99	1.00	0.99	66
4	1.00	0.98	0.99	54
5	0.98	1.00	0.99	55
6	0.98	0.96	0.97	53
...		*see remaining classes*		

Table 11: Classification report for Y_3: *FaultCategory* (remaining).

Class	Precision	Recall	F1-Score	Support
7	1.00	1.00	1.00	67
8	1.00	1.00	1.00	57
Accuracy		0.989 (on 467 samples)		
Macro Avg	0.99	0.95	0.97	467
Weighted Avg	0.99	0.99	0.99	467

6.3. Overall Performance Comparison

We compare NRP-GCN-RF with SVM (one-hot) and RF (tabular). As shown in Table 12, the proposed model consistently outperforms baselines across tasks, with notable gains in recall/F1 for Y_1/Y_2.

Table 12: Overall performance across three tasks (best in **bold**).

Model	Y_1: HasEmergencyFee			Y_2: EmergencyFeeCategory			Y_3: FaultCategory		
	Precision	Recall	F1-Score	Precision	Recall	F1-Score	Precision	Recall	F1-Score
SVM	0.74	0.74	0.74	0.74	0.72	0.73	0.80	0.80	0.79
RF	0.90	0.90	0.90	0.92	0.92	0.92	0.95	0.95	0.95
NRP-GCN-RF	**0.94**	**0.94**	**0.94**	**0.93**	**0.93**	**0.93**	**0.98**	**0.99**	**0.99**

Key insight. By leveraging graph structure with a decoupled RF head, NRP-GCN-RF improves detection of high-cost and minority scenarios without sacrificing overall calibration (Appendix).

6.4. Cross-Validation Performance

Table 13 reports fold-wise accuracies, indicating stable generalization under vehicle-wise chronological splits.

Table 13: Cross-validation accuracy across tasks with NRP-GCN-RF.

Task	Fold Accuracies	Mean Accuracy	Std. Dev.
Y_1: HasEmergencyFee	0.9130, 0.9565, 0.8913, 0.9333, 0.8889	0.9166	0.0257
Y_2: EmergencyFeeCategory	0.9690, 0.9297, 0.9609, 0.9844, 0.9375	0.9563	0.0202
Y_3: FaultCategory	0.9968, 0.9838, 0.9903, 0.9870, 0.9773	0.9870	0.0065

Summary. Across tasks, NRP-GCN-RF achieves high accuracy with low variance, strong sensitivity to costly events, and interpretable associations that assist downstream diagnostics. Additional robustness, calibration, and efficiency analyses are provided in Appendix.

6.5. Fault Code Combination Analysis

Protocol. Feature contributions are estimated by *permutation importance* on the RF head using out-of-fold predictions under the vehicle-wise, chronological CV protocol (Sec. 6). Candidate inputs include graph-aggregated operational features (`Count`, `Proportion`), geospatial descriptors (`AvgLatitude`, `AvgLongitude`, `LatitudeDiff`, `LongitudeDiff`), and Apriori-stable code-combination indicators (e.g., `Combination_6699-19`, `Combination_609-14`, `Combination_444-12`, `Combination_100-18`) . Importances are min–max normalized per task and summarized by the median across folds.

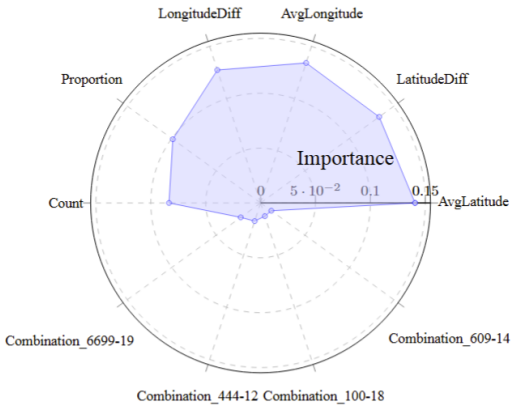


Figure 9: Feature-importance radar. Normalized permutation importance for representative features across tasks (Y_1: *HasEmergencyFee*, Y_2: *EmergencyFeeCategory*, Y_3: *FaultCategory*). Geospatial (`AvgLatitude`, `AvgLongitude`, `LatitudeDiff`, `LongitudeDiff`), operational (`Proportion`, `Count`), and Apriori-informed combinations (`Combination_*`) are included.

Results. Figure 9 shows a geospatially dominated profile: `AvgLatitude` attains the largest normalized contribution (≈ 0.15), followed by `LatitudeDiff` and `AvgLongitude`. Operational intensity features (`Proportion`, `Count`) rank mid-range. In contrast, individual

combination flags (`Combination_6699-19`, `Combination_609-14`, `Combination_444-12`, `Combination_100-18`) exhibit comparatively lower direct importance. This is consistent with Sec. 6.1: frequent/high-lift pairs are informative primarily through graph construction and message passing, while coarse geospatial and operational aggregates dominate the final decision for Y_1/Y_2 and support Y_3.

Learning and capacity. Learning/validation curves are reported in Fig. 10. The top row increases the training-set size; the bottom row varies the number of trees in the RF head. Solid lines denote mean accuracy across folds and shaded regions indicate the interquartile range (IQR).

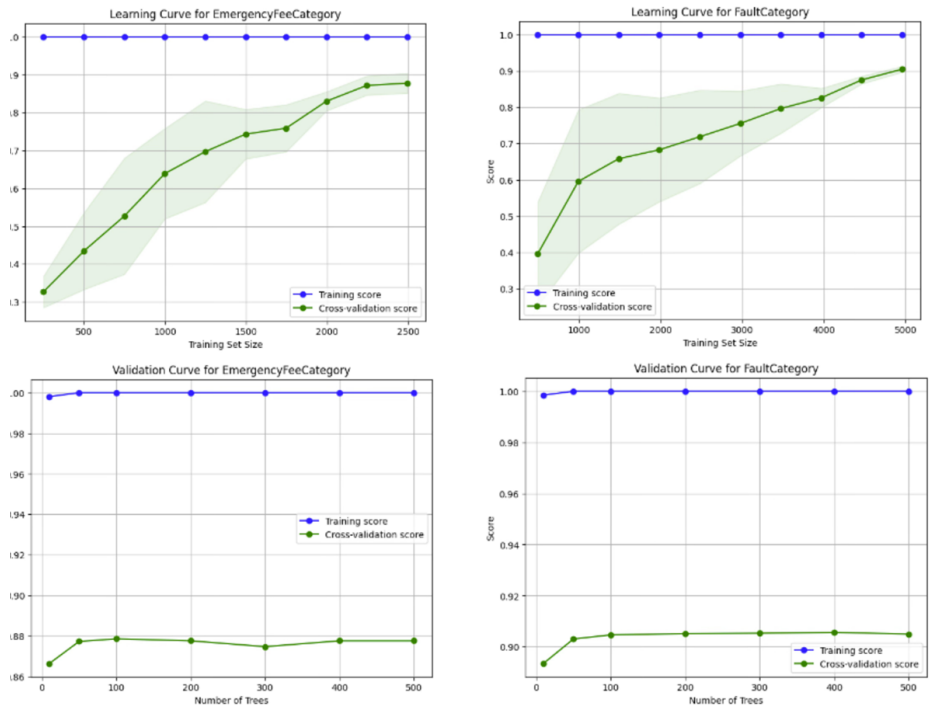


Figure 10: Learning/validation curves for Y_2 and Y_3 under vehicle-wise chronological CV. Top: accuracy vs. training-set size (means with IQR). Bottom: accuracy vs. number of trees.

Analysis. For Y_2, validation accuracy keeps improving with larger training sets and deeper trees. For Y_3, accuracy plateaus earlier, indicating near-saturation under current features. The training–validation gap suggests more diverse data or regularization would help.

Implications. Geospatial descriptors (`AvgLatitude`, `LatitudeDiff`, `AvgLongitude`) are the principal drivers across tasks; operational aggregates provide secondary gains. Explicit code-combination indicators contribute complementary signal but are most effective when realized through the graph's multi-factor edges rather than as stand-alone tabular features. Consequently, for Y_2 (cost prediction), broadening route/depot coverage and enriching location metadata are expected to yield the largest improvements, while Y_3 (fault categorization) appears near saturation under the current feature set.

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7. Conclusion and Future Work

This paper presents **NRP-GCN-RF**, a hybrid framework that integrates graph neural networks with ensemble learning for predictive maintenance in commercial vehicle fleets. By transforming chassis-level fault logs into relational graphs and incorporating operational features, the model captures both structural dependencies and statistical characteristics of real-world maintenance data. Evaluated across three prediction tasks—*HasEmergencyFee*, *EmergencyFeeCategory*, and *FaultCategory*—NRP-GCN-RF consistently outperforms baseline methods, particularly under class imbalance and cost-sensitive scenarios. In parallel, association rule mining using the Apriori algorithm reveals frequent co-fault patterns, enhancing interpretability and supporting subsystem-level diagnostics. Together, these components form a unified and explainable solution for failure forecasting and maintenance planning.

Future work will explore three directions. First, we will extend the framework toward modeling temporal graph dynamics, enabling the capture of evolving fault propagation and degradation patterns over continuous maintenance cycles. Second, we will investigate interpretable graph reasoning by incorporating attention mechanisms and explainable GNN modules to highlight critical nodes, edges, and motifs in predictive decisions. Third, we aim to enhance cross-fleet generalization by applying the framework to multi-fleet or multi-vendor datasets, fusing SPN–FMI codes with real-time telemetry and contextual metadata to broaden applicability.

Advancing along these directions will improve adaptability, transparency, and cost-awareness in predictive maintenance systems, enabling more reliable and scalable deployment in industrial vehicle operations.

8. Acknowledgement

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